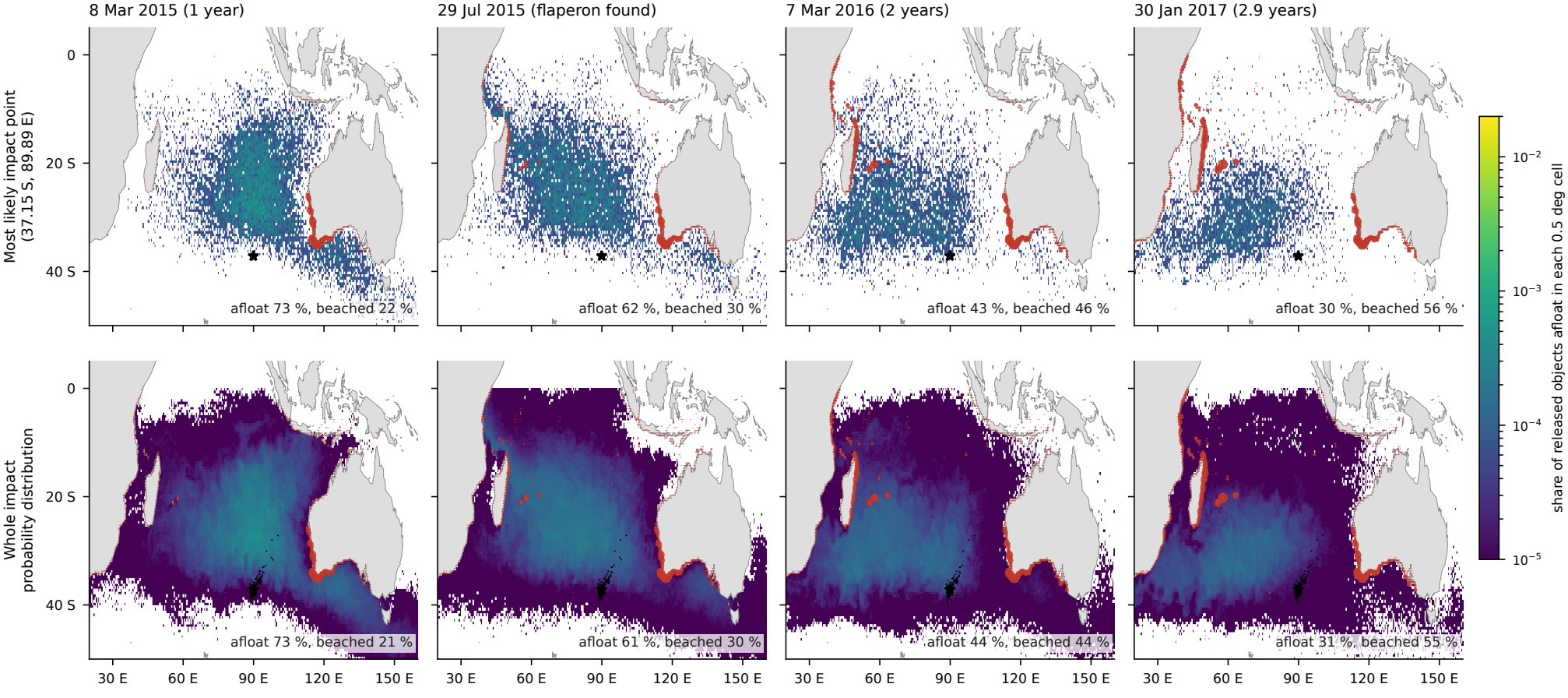
Where floating debris goes from the most probable impact area (about 37 S), March 2014 to January 2017: two ocean models

Colour: objects still afloat. Red dots: where objects have come ashore so far (dot area = share). Star: most likely impact point. Black dots (lower row): 200 start points drawn from the impact distribution.

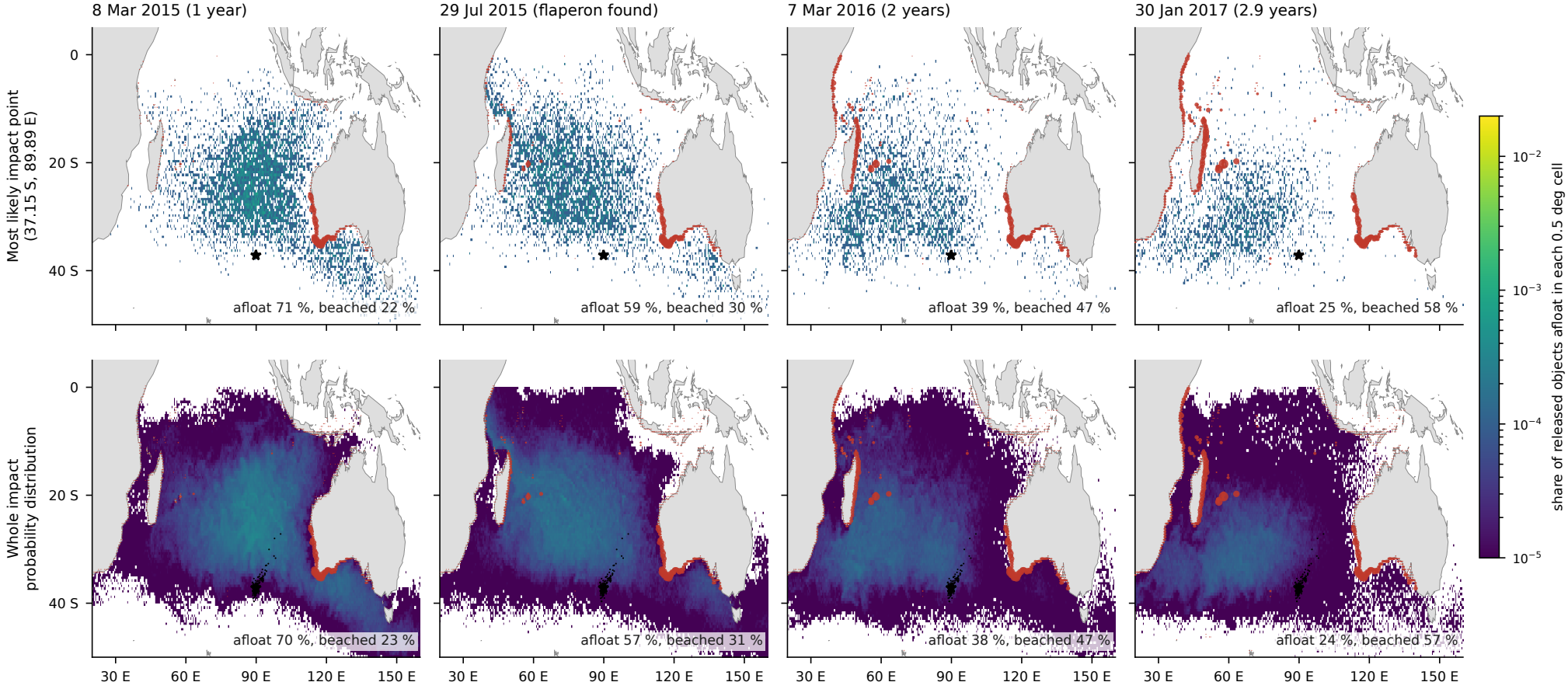


Plain English. The start points come only from the flight-path calculation (satellite data and aircraft performance); the debris finds are not used to choose them. The upper row starts all objects at the most likely impact point. The lower row starts objects at 200 points drawn at random from the impact probability distribution, so it shows the spread that comes from the uncertainty in the impact point. Each start point releases about 10,000 floating objects of three types (flaperon-like, low-exposure exterior parts, high-windage interior parts) in four ocean realisations. The objects drift with the ocean current, with part of the wind and with small random eddies. They stop when they reach a coast and do not float off again. The model tracks 15-160 E, 50 S to the equator; objects that leave it are no longer shown. Two ocean models (GLORYS12 and GlobCurrent) are given equal weight.

Technical. Impact PDF: run J waypoints end-of-flight posterior, 00:19 option D0' (00:19 data held out), cell 'D0 total q=0.333 (C2 silent)', seeds 1-8 of 24 pooled with weights proportional to seed evidence (declared subset); drift likelihood not applied. Draws: 200 weighted samples, latitude 5/50/95 % -38.52/-37.15/-34.05 (full 24-seed table -38.40/-37.12/-33.87). Mode on a 0.05 deg grid. Drift: untracked stand-in harness `snapshots` on drift d51d80a (OceanSetup::run_tracks_ep, class_responses, diffusivity_prior, ocean error per environment seed 1-4), 15-160 E, GSHHG 2.3.7 coast, snap 25 km, refloat off, 6 h step, K log-uniform 100-1000 m2/s, 834 particles per class per seed, no importance splitting. GLORYS12V1 0.494 m + ERA5; GlobCurrent 0 m + ERA5 with product-relative windage U(-0.75,-0.60) % of U10. Release 2014-03-08 00:19:37 UTC. Shares = count / released, equal weight per ocean model.

Where floating debris goes from the most probable impact area (about 37 S): GLORYS12 ocean model only

Colour: objects still afloat. Red dots: where objects have come ashore so far (dot area = share). Star: most likely impact point. Black dots (lower row): 200 start points drawn from the impact distribution.

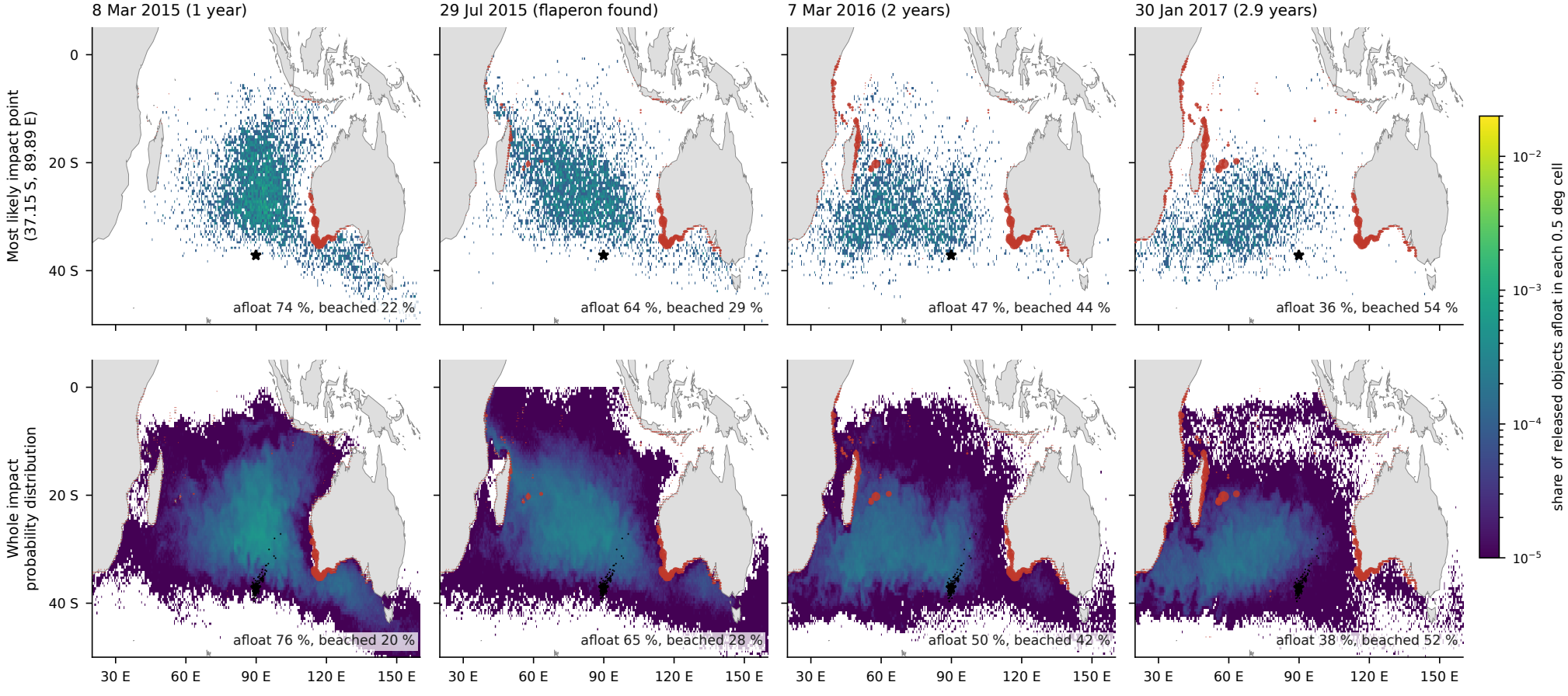


Plain English. The start points come only from the flight-path calculation (satellite data and aircraft performance); the debris finds are not used to choose them. The upper row starts all objects at the most likely impact point. The lower row starts objects at 200 points drawn at random from the impact probability distribution, so it shows the spread that comes from the uncertainty in the impact point. Each start point releases about 10,000 floating objects of three types (flaperon-like, low-exposure exterior parts, high-windage interior parts) in four ocean realisations. The objects drift with the ocean current, with part of the wind and with small random eddies. They stop when they reach a coast and do not float off again. The model tracks 15-160 E, 50 S to the equator; objects that leave it are no longer shown. GLORYS12 ocean model only.

Technical. Impact PDF: run J waypoints end-of-flight posterior, 00:19 option D0' (00:19 data held out), cell 'D0 total q=0.333 (C2 silent)', seeds 1-8 of 24 pooled with weights proportional to seed evidence (declared subset); drift likelihood not applied. Draws: 200 weighted samples, latitude 5/50/95 % -38.52/-37.15/-34.05 (full 24-seed table -38.40/-37.12/-33.87). Mode on a 0.05 deg grid. Drift: untracked stand-in harness `snapshots` on drift d51d80a (OceanSetup::run_tracks_ep, class_responses, diffusivity_prior, ocean error per environment seed 1-4), 15-160 E, GSHHG 2.3.7 coast, snap 25 km, refloat off, 6 h step, K log-uniform 100-1000 m2/s, 834 particles per class per seed, no importance splitting. GLORYS12V1 0.494 m + ERA5; GlobCurrent 0 m + ERA5 with product-relative windage U(-0.75,-0.60) % of U10. Release 2014-03-08 00:19:37 UTC. Shares = count / released, equal weight per ocean model.

Where floating debris goes from the most probable impact area (about 37 S): GlobCurrent ocean model only

Colour: objects still afloat. Red dots: where objects have come ashore so far (dot area = share). Star: most likely impact point. Black dots (lower row): 200 start points drawn from the impact distribution.



Plain English. The start points come only from the flight-path calculation (satellite data and aircraft performance); the debris finds are not used to choose them. The upper row starts all objects at the most likely impact point. The lower row starts objects at 200 points drawn at random from the impact probability distribution, so it shows the spread that comes from the uncertainty in the impact point. Each start point releases about 10,000 floating objects of three types (flaperon-like, low-exposure exterior parts, high-windage interior parts) in four ocean realisations. The objects drift with the ocean current, with part of the wind and with small random eddies. They stop when they reach a coast and do not float off again. The model tracks 15-160 E, 50 S to the equator; objects that leave it are no longer shown. GlobCurrent ocean model only.

Technical. Impact PDF: run J waypoints end-of-flight posterior, 00:19 option D0' (00:19 data held out), cell 'D0 total q=0.333 (C2 silent)', seeds 1-8 of 24 pooled with weights proportional to seed evidence (declared subset); drift likelihood not applied. Draws: 200 weighted samples, latitude 5/50/95 % -38.52/-37.15/-34.05 (full 24-seed table -38.40/-37.12/-33.87). Mode on a 0.05 deg grid. Drift: untracked stand-in harness `snapshots` on drift d51d80a (OceanSetup::run_tracks_ep, class_responses, diffusivity_prior, ocean error per environment seed 1-4), 15-160 E, GSHHG 2.3.7 coast, snap 25 km, refloat off, 6 h step, K log-uniform 100-1000 m2/s, 834 particles per class per seed, no importance splitting. GLORYS12V1 0.494 m + ERA5; GlobCurrent 0 m + ERA5 with product-relative windage U(-0.75,-0.60) % of U10. Release 2014-03-08 00:19:37 UTC. Shares = count / released, equal weight per ocean model.